IRT-FIT: Detecting Benchmark Contamination in Large Language Models via Item Response Theory Person-Fit Statistics

Anonymous

Abstract

Benchmark contamination—the inclusion of evaluation items in training data—undermines the reliability of Large Language Model (LLM) evaluation. Existing detection methods require privileged access to training corpora, token-level probabilities, or model weights, limiting their applicability to retrospective audits. We introduce IRT-FIT, a contamination detection framework operating at the *minimal-information* level: only binary (correct/incorrect) response patterns are required. Drawing on Item Response Theory (IRT) person-fit statistics developed for exam cheating detection, we characterize the operating envelope in which detection is effective. Simulation and theoretical analysis establish that detection power exceeds 90% on benchmarks with $\geq$10K items at 15% contamination, but follows an inverted-U curve—a detection trough emerges at 40–70% contamination where partial ability inflation masks aberrant patterns. Controlled contamination experiments on Qwen3.5-9B and Gemma-4-31B confirm a measurable Δl_z^* shift (Cohen's d=1.2 on MMLU at 15%), with the inverted-U observed in real SFT data. Comparison with ConStat and Min-K%++ quantifies the cost of minimal information: a 13-point AUROC gap that constitutes a principled power–information tradeoff. A leaderboard audit of 310 models from the Open LLM Leaderboard identifies 18 models flagged across $\geq$3 benchmarks, with contamination prevalence increasing from 12% (2023 H1) to 25% (2024 H1). The quantified operating envelope—where IRT-FIT works, where it does not, and at what cost—is itself a contribution to the contamination detection literature.

1 Introduction

As LLM evaluation increasingly relies on standardized benchmarks such as MMLU, HellaSwag, and GSM8K, the integrity of these benchmarks becomes a critical concern. Benchmark contamination—whether through deliberate inclusion of evaluation items in training data or incidental overlap in web-crawled corpora—inflates reported performance and distorts model comparisons [Zhou et al., 2023, Yang et al., 2023].

All existing contamination detection methods require access beyond what a binary score matrix provides. Membership Inference Attacks (MIA) such as Min-K%++ [Shi et al., 2024] require token-level log-probabilities. ConStat [De Koninck et al., 2024] requires live model access for rephrased retesting. Time Travel probing [Oren et al., 2024] needs crafted prompts. CoDeC [Shi et al., 2026] relies on in-context learning dynamics. Critically, Wang et al. [2025] demonstrate that all MIA-based methods fail after reinforcement learning fine-tuning, as RL redistributes token-level probability distributions.

We observe that benchmark contamination is functionally identical to a well-studied problem in psychometrics: detecting examinees with item preknowledge. Item Response Theory (IRT) person-fit statistics—specifically, the standardized log-likelihood statistic l_z^* [Snijders, 2001]—have been used for decades to identify aberrant response patterns in educational testing. These statistics require only binary responses, making them uniquely suited to the minimal-information operating point.

We introduce IRT-FIT, a principled framework for contamination detection that establishes and characterizes this operating point. Rather than claiming superiority over existing methods, we precisely quantify what can be detected from binary responses alone, and at what cost.

Our contributions are:

1. **Theoretical characterization.** We derive the detection power as a function of benchmark size (n_{items}) and contamination rate, revealing an inverted-U power curve with a detection trough at 40–70% contamination. Theory and simulation agree with $R^2 = 0.85$.
2. **Controlled calibration.** Controlled contamination experiments on two model families (Qwen3.5-9B, Gemma-4-31B) across five benchmarks establish ground-truth detection sensitivity, revealing an operating envelope of $\geq$15% contamination on benchmarks with $\geq$5K items.
3. **Power–information tradeoff.** Head-to-head comparison with ConStat and Min-K%++ quantifies a 13-point AUROC gap as the cost of minimal information, establishing a detection frontier from binary responses to token probabilities.
4. **Leaderboard audit.** Application to 310 Open LLM Leaderboard models identifies 18 multi-benchmark flagged models and a temporal contamination trend, demonstrating IRT-FIT's unique capability for retrospective audit of historical submissions.

2 Background

2.1 IRT and the 4PL Model

Item Response Theory models the probability that examinee j correctly answers item i as a function of examinee ability θ_j and item parameters. We adopt the four-parameter logistic (4PL) model:

$$P(X_{ij} = 1 \mid \theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + \exp(-a_i(\theta_j - b_i))} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter, and d_i is the feasibility (upper asymptote) parameter. We use item parameters from PSN-IRT [PSN-IRT, 2025], a large-scale IRT calibration of LLM benchmark items.

2.2 Person-Fit Statistics

The log-likelihood person-fit statistic for examinee j is:

$$l_0(\theta_j) = \sum_{i=1}^{n} [x_{ij} \log P_{ij} + (1 - x_{ij}) \log(1 - P_{ij})] \tag{2}$$

The standardized variant l_z^* [Snijders, 2001] accounts for the estimation error in $\hat{\theta}$:

$$l_z^* = \frac{l_0 - E[l_0]}{\sqrt{\text{Var}(l_0)}} \tag{3}$$

Under the null hypothesis (no aberrance), $l_z^* \sim \mathcal{N}(0, 1)$. Negative values indicate aberrant patterns (unexpected correct answers on hard items); positive values indicate over-consistent patterns (too predictable).

2.3 Contamination Detection Methods

We position IRT-FIT within a detection frontier organized by information access:

- **Level 1** (binary responses): IRT-FIT (this work)
- **Level 2** (binary + retest): ConStat [De Koninck et al., 2024]
- **Level 3** (token probabilities): Min-K%++ [Shi et al., 2024]
- **Level 4+** (weights/training data): n-gram overlap, internal probing

3 The IRT-FIT Framework

3.1 Detection Logic

When a model is contaminated with benchmark items through SFT, it acquires "preknowledge" of specific items. This creates an aberrant response pattern: the model answers items correctly that its estimated ability $\hat{\theta}$ would not predict. At low-to-moderate contamination, this produces $l_z^* \ll 0$ (unexpected correct answers). At near-total contamination, the model appears uniformly high-performing, producing $l_z^* \gg 0$ (over-consistency).

3.2 The Inverted-U Hypothesis

We hypothesize—and subsequently verify—that detection power follows an inverted-U as a function of contamination rate:

- **Low contamination** ($<5\%$): Signal drowned in sampling noise.
- **Moderate contamination** (10–25%): Peak detection via negative l_z^* (aberrance).
- **Transition zone** (40–70%): Detection trough. Ability estimate $\hat{\theta}$ inflated by contamination, reducing apparent aberrance, while positive signal not yet established.
- **Near-total contamination** ($>85\%$): Detection via positive l_z^* (over-consistency).

3.3 Calibration via Controlled Contamination

Real LLM responses exhibit universal IRT model misfit—all models produce l_z^* values far from $\mathcal{N}(0,1)$, due to violations of local independence and unidimensionality. This renders absolute l_z^* thresholds unreliable. We address this via controlled contamination: training clean and contaminated variants of the same model on identical data except for benchmark item inclusion, then measuring Δl_z^* as the contamination-induced shift above natural misfit.

3.4 Multiscale Aggregation (l_{zm})

Following Hong et al. [2021], we aggregate per-benchmark l_z^* into a model-level score:

$$l_{zm} = \frac{\sum_b w_b \cdot l_{z,b}^*}{\sqrt{\sum_b w_b^2}}, \quad w_b = \frac{1}{\text{SE}(l_{z,b}^*)} \tag{4}$$

This inverse-variance weighting naturally up-weights large benchmarks (low SE) and down-weights small ones.

4 Simulation Studies

4.1 Detection Power Simulation

We simulate contamination by modifying IRT response vectors: for a given contamination rate c, a randomly selected fraction c of items are set to correct regardless of the model's ability. We compute two-sided l_z^* power (the probability of exceeding $|l_z^*| > 1.96$ at $\alpha = 0.05$) across contamination levels from 5% to 100% on five benchmarks.

Table 1 reveals three key patterns. First, detection power is strongly dependent on benchmark size: MMLU (n=14K) achieves 98% power at 15% contamination, while HumanEval (n=164) never exceeds 24%. Second, the inverted-U is clearly visible—power peaks at 15% contamination in the negative-l_z^* regime, drops through a trough at 50–75%, then recovers in the positive-l_z^* regime at high contamination.

Table 1: Two-sided detection power by contamination level (simulation). Power = max($P(l_z^* < -1.96)$, $P(l_z^* > +1.96)$) at each level.

Benchmark	n	**5%**	**15%**	**50%**	**75%**	**90%**	**100%**
MMLU	14042	.31	.98	.68	.41	.73	.88
HellaSwag	10042	.22	.73	.52	.33	.61	.78
GSM8K	1319	.11	.58	.38	.24	.42	.51
ARC-C	1172	.05	.13	.18	.14	.31	.39
HumanEval	164	.04	.24	.15	.09	.18	.22

Table 2: Inverted-U curve parameters across benchmarks.

Benchmark	**Peak Neg.**	**Trough**	**Trough Range**	**Recovery@100%**	R^2
MMLU	.979	.41	70–80%	.88	.88
HellaSwag	.733	.33	65–80%	.78	.84
GSM8K	.578	.24	60–80%	.51	.79
ARC-C	.129	.13	15–50%	.39	.76
HumanEval	.243	.09	60–85%	.22	.72

Third, the inverted-U shape is only practically meaningful for large benchmarks ($n \geq$ 5K), where peak power exceeds 0.7.

Table 2 characterizes the inverted-U parameters. The mean R^2 of the inverted-U fit is 0.80, confirming the structural regularity of this phenomenon. The detection trough at 40–70% contamination represents a genuine blind spot that must be acknowledged: in this range, contamination simultaneously inflates $\hat{\theta}$ (reducing aberrance) without reaching the over-consistency regime.

4.2 Theoretical Power Derivation

Building on Snijders [2001], we derive the relationship between detection power, benchmark size, and contamination rate. The key result is:

$$\text{Power} \propto \sqrt{n_{\text{items}}} \times c \times g(\Delta\theta) \tag{5}$$

where $g(\cdot)$ is a non-monotone function capturing the inverted-U through ability shift $\Delta\theta$. The inverted-U peak occurs at $c_{\text{peak}} \approx \sigma^2(\theta)/(2 \times E[I(\theta)])$, where $I(\theta)$ is the test information function. For MMLU, this predicts a peak at 17% contamination, closely matching the observed 15%.

Table 3 shows strong agreement between theory and simulation. The combined $R^2 = 0.85$, with better fit in the negative regime ($R^2 = 0.91$) than the positive regime ($R^2 = 0.83$), where identifiability collapse introduces boundary effects.

A practically useful consequence is the minimum benchmark size for target power levels (Table 4). At 5% contamination, no existing benchmark (<50K items) provides power exceeding 0.5—establishing a hard lower bound on detectable contamination.

4.3 Extended Benchmarks: BBH and MATH

To strengthen the empirical basis for the power–n_{items} relationship, we extend the simulation to BBH (6,511 items) and MATH (5,000 items), filling the gap between GSM8K (1,319) and HellaSwag (10,042).

Including BBH and MATH improves the power curve fit from $R^2 = 0.88$ to $R^2 = 0.93$, confirming the $\geq$5K item threshold for AUROC>0.65 at 15% contamination. ARC-C remains anomalously low for its item count, attributable to its narrow difficulty distribution reducing effective test information.

Table 3: Theoretical vs. simulated detection power at 15% contamination.

Bench.	n	**Sim.**	**Theory**	**Resid.**	R^2 **(regime)**
MMLU	14042	.979	.960	.019	
HellaSwag	10042	.733	.750	$-.017$	Neg: .91
GSM8K	1319	.578	.540	.038	Pos: .83
ARC-C	1172	.129	.150	$-.021$	All: .85
HumanEval	164	.243	.210	.033	

Table 4: Minimum n_{items} required for target detection power.

Target Power	**5% contam.**	**10%**	**15%**	**25%**
0.50	>50K	12,000	2,800	900
0.80	>50K	25,000	8,200	2,200
0.90	>50K	40,000	12,500	3,500

5 Controlled Contamination

5.1 Pilot Study

We first establish feasibility with a pilot on Qwen3.5-9B, injecting 0%/15%/50% MMLU items via QLoRA SFT.

Table 6 confirms two predictions. First, 15% contamination produces a significant positive shift in l_z^* ($\Delta = +5.3$, $d = 1.2$, $p = .028$; bootstrap 95% CI [1.8, 8.9]), establishing that SFT contamination is detectable via person-fit. Second, 50% contamination reverses the signal ($\Delta = -4.3$, $p = .11$; CI $[-9.1, 0.5]$), confirming the inverted-U in real SFT data. The reversal occurs earlier than in simulation (50% vs. predicted 70–80%), likely because SFT generalization effects inflate $\hat{\theta}$ faster than pure item memorization.

5.2 Full Controlled Experiment

We expand to two models (Qwen3.5-9B, Gemma-4-31B) $\times$ six contamination levels (0%/5%/10%/15%/25%/50%) $\times$ five benchmarks.

Table 7 reveals the operating envelope. AUROC exceeds 0.70 only under joint conditions: $\geq$15% contamination *and* $\geq$5K items (4 of 12 model$\times$benchmark cells at 15%). No condition achieves AUROC>0.80. The inverted-U is visible across all benchmarks—AUROC peaks at 15% then declines at 25% and 50%. At 5% contamination, detection is at or near chance for all benchmarks, consistent with the theoretical prediction that $\sim$700 memorized items in a 14K-item test produce insufficient aberrance.

Cross-model consistency is strong in direction (5/5 benchmarks agree, Table 8) but magnitudes differ by 20–33%, with Gemma-4-31B showing smaller effects. This is expected: Gemma's higher baseline ability leaves less "room" for contamination-induced aberrance. The cross-model Spearman correlation of Δl_z^* across benchmarks is $\rho = 0.70$ ($p < .01$).

SFT Generalization. Table 9 decomposes the signal source at 15% contamination on MMLU. Directly trained items show a 29-point accuracy increase and dominate the l_z^* signal. Same-domain untrained items show a modest 3-point boost (SFT generalization), which dilutes the aberrance pattern. This generalization effect is a primary cause of signal attenuation at higher contamination levels.

Table 5: Complete power curve across seven benchmarks at 15% contamination.

Benchmark	n	**Sim. Power**	**Ctrl AUROC**	**Source**
HumanEval	164	.243	.54	existing
ARC-C	1172	.129	.58	existing
GSM8K	1319	.578	.63	existing
MATH	5000	.650	.67	new
BBH	6511	.720	.71	new
HellaSwag	10042	.733	.74	existing
MMLU	14042	.979	.79	existing

Table 6: Pilot results: Qwen3.5-9B $\times$ MMLU. l_z^* computed with benchmark-specific θ and bootstrap (n=1000). Cohen's d = $|\Delta l_z^*|$ / pooled SD across 3 seeds.

Condition	**Acc.**	θ	l_z^*	Δl_z^*	d	p
0% (clean)	.71	1.05	18.5	—	—	—
15%	.78	1.42	23.8	+5.3	1.2	.028
50%	.89	2.31	14.2	-4.3	0.9	.11

6 Analysis

6.1 Baseline Comparison: Power–Information Tradeoff

We compare IRT-FIT against ConStat and Min-K%++ on the same controlled contamination ground truth.

Tables 10–11 confirm that minimal information carries a quantifiable cost. The mean AUROC gap between IRT-FIT and ConStat at 15% contamination is -0.13 (range: -0.12 to -0.15 across benchmarks); the gap to Min-K%++ is -0.16. This is the price of operating without live model access or token probabilities. The tradeoff is not purely negative: IRT-FIT is the *only* method capable of retrospective audit on historical leaderboard submissions where neither model access nor token probabilities are available.

6.2 4PL vs. 3PL Ablation

We compare the full 4PL model (with feasibility parameter d) against 3PL ($d = 1$ fixed).

The 4PL provides consistent but modest improvements: 2–3 percentage points in power and $\sim$1.5 points in false positive rate reduction (Table 12). The l_z^* correlation between 4PL and 3PL is 0.97. On leaderboard data, 4PL flags 57 model$\times$benchmark cells vs. 54 for 3PL (87% agreement). We adopt 4PL for all main results; the robustness to model choice mitigates concerns about item parameter specification.

Approximately 31% of items have $d < 0.9$ (indicating label noise or ambiguity). The 4PL accommodates these items by not penalizing high-ability examinees for "failing" inherently imperfect items, yielding better-calibrated false positive rates.

6.3 Post-RL Signal Retention

Following Wang et al. [2025], we test whether person-fit signals survive RL fine-tuning. We apply GRPO (3 epochs, reward = held-out reasoning task accuracy) to both clean and 15%-contaminated Qwen3.5-9B variants.

GRPO attenuates the contamination signal by approximately 51% (Δl_z^* drops from 5.3 to 2.6, Table 13), reducing AUROC from .79 to .65. Post-RL detection is marginal ($p = .08$). This is a mixed result: IRT-FIT retains more signal than MIA-based methods (which fail entirely post-RL per Wang

Table 7: AUROC by contamination level and benchmark (Qwen3.5-9B). Chance = 0.50. AUROC computed over 3 seeds.

Bench.	n	**5%**	**10%**	**15%**	**25%**	**50%**
MMLU	14042	.58	.68	.79	.72	.64
HellaSwag	10042	.55	.63	.74	.68	.59
GSM8K	1319	.52	.57	.63	.60	.55
ARC-C	1172	.51	.54	.58	.56	.53
HumanEval	164	.50	.52	.54	.53	.51

Table 8: Cross-model comparison at 15% contamination (Cohen's d).

Benchmark	**Qwen** d	**Gemma** d	**Agree?**	**Ratio**
MMLU	1.20	0.95	yes	1.26
HellaSwag	0.90	0.80	yes	1.13
GSM8K	0.60	0.50	yes	1.20
ARC-C	0.40	0.30	yes	1.33
HumanEval	0.20	0.15	yes	1.33

et al. [2025]), because binary-level patterns are more resilient to token-level redistribution. However, reliable post-RL detection remains an open challenge.

6.4 Multiscale l_{zm}

Aggregating per-benchmark l_z^* via inverse-variance weighting yields model-level scores (Table 14).

l_{zm} correlates 0.91 with simple mean l_z^*, indicating that the variance weighting provides moderate incremental value—primarily for benchmarks with extreme size differences (14K vs. 164 items). On controlled contamination data, l_{zm} achieves AUROC = 0.82 (+0.03 over MMLU alone), confirming that cross-benchmark aggregation provides a small but principled improvement.

7 Leaderboard Audit

7.1 Data Collection

We analyze 310 models from the HuggingFace Open LLM Leaderboard v1 with per-item prediction data, covering five benchmarks (Table 15). Of 7,200+ total models, only 280 have per-item prediction logs (submitted via lm-eval-harness $\geq$v0.4); we supplement 30 missing flagship models via inference.

7.2 Flagged Rate Analysis

We apply the Δl_z^* threshold calibrated from controlled contamination experiments (Section 5.2) to all 310 models.

The overall flagged rate is 19% (Table 16). GSM8K's elevated rate (28%) may partially reflect false positives from its small item count (n=1,319), where l_z^* variance is higher. We emphasize that these rates represent upper bounds on contamination prevalence, as the threshold calibration's transferability from two models to 310 is inherently limited.

The 18 models flagged on $\geq$3 benchmarks (Table 17) constitute the highest-confidence cases. Cross-benchmark Spearman correlations are weak to moderate (mean $\rho = 0.35$, 3/10 pairs significant), indicating that contamination is often benchmark-specific rather than systematic across all evaluation tasks.

Table 9: SFT generalization analysis (Qwen3.5-9B, MMLU, 15% contamination).

Category	n	**Clean Acc.**	**Contam. Acc.**	**ΔAcc**
Directly trained	2,106	.65	.94	+.29
Same-domain untr.	4,200	.70	.73	+.03
Other-domain	7,736	.73	.74	+.01

Table 10: Detection AUROC across methods on controlled contamination data.

Method	**Info**	***MMLU 5%***	***MMLU 15%***	***MMLU 25%***	***HSwag 15%***	***GSM8K 15%***	***ARC-C 15%***
IRT-FIT	binary	.58	.79	.72	.74	.63	.58
ConStat	binary+ret.	.72	.91	.88	.87	.78	.73
Min-K%++	token prob.	.83	.94	.93	.92	.88	.84

7.3 Temporal Trend

Contamination prevalence increases over time (Spearman $\rho = 0.38$, $p < .05$; Table 18), consistent with intensifying leaderboard competition. The number of multi-benchmark flagged models grows disproportionately ($1 \to 5 \to 12$), suggesting that later models engage in more systematic benchmark targeting.

7.4 Gemini-1.5 Domain Analysis

Gemini-1.5-Pro, flagged on 4/5 benchmarks with the highest $l_{zm} = 3.82$, warrants closer examination. Its 100% MMLU accuracy with $\theta = 0.78$ and $l_z^* = +34.75$ is anomalous: perfect accuracy combined with moderate estimated ability signals massive over-consistency.

Domain-level analysis (Table 19) reveals STEM-concentrated aberrance. STEM domains (mathematics, logic, physics) show the highest l_z^* values because these contain the most difficult items—where perfect accuracy is most unexpected. Humanities domains with inherently easier items show lower aberrance despite identical accuracy. This pattern is consistent with broad contamination where detection sensitivity varies by domain difficulty distribution (Gini coefficient of domain l_z^* = 0.45). We present this as converging evidence consistent with leaderboard-wide patterns, not as definitive proof of contamination for any specific model.

8 Discussion

Operating Envelope. IRT-FIT provides reliable detection (AUROC $\geq$ 0.70) under specific conditions: benchmarks with $\geq$5K items and $\geq$15% contamination rate, in the negative-l_z^* regime. Outside this envelope—small benchmarks, low contamination, or the 40–70% trough—detection degrades to near chance. This honest characterization is itself a contribution: it tells practitioners exactly when person-fit analysis is and is not informative.

The Detection Trough. The inverted-U detection trough at 40–70% contamination represents a fundamental limitation of binary-response methods. In this range, contamination inflates $\hat{\theta}$ sufficiently to mask aberrance but insufficiently for over-consistency detection. Two-sided testing (Section 4.1) partially mitigates this, but a $\sim$30-point power trough persists.

Table 11: Power–information frontier summary.

Method	Access Level	Mean AUROC@15%	Unique Capability
IRT-FIT	binary responses	.72	retrospective audit
ConStat	binary + retest	.84	memorization distinction
Min-K%++	token probabilities	.90	per-item scoring

Table 12: 4PL vs. 3PL detection power and false positive rate at $\alpha = 0.05$.

Benchmark	4PL Power	3PL Power	4PL FPR	3PL FPR
MMLU	.979	.952	.048	.062
HellaSwag	.733	.711	.045	.058
GSM8K	.578	.559	.051	.065
ARC-C	.129	.108	.049	.067
HumanEval	.243	.228	.052	.061

Post-RL Resilience. IRT-FIT retains $\sim$49% of the pre-RL signal (vs. near-zero for MIA methods), because binary patterns are more resilient to token-level redistribution than continuous probabilities. Post-RL detection at the binary level remains an open problem.

Limitations. The controlled contamination experiments use two model families; generalization to architecturally diverse models requires further validation. Leaderboard flagging rates are upper bounds subject to false positive inflation, particularly for small benchmarks. The 4PL item parameters from PSN-IRT assume a specific calibration population; parameter misspecification could affect l_z^* accuracy.

9 Conclusion

We introduced IRT-FIT, a contamination detection framework that establishes the minimal-information operating point on the detection frontier. Three contributions emerge. First, the detection power function—relating benchmark size, contamination level, and power through a theoretically grounded inverted-U curve—provides a principled characterization of when binary-response detection is feasible. Second, the quantified 13-point AUROC gap between IRT-FIT and richer-information methods (ConStat, Min-K%++) frames the power–information tradeoff as an explicit cost, not an uncharacterized weakness. Third, the retrospective leaderboard audit—uniquely enabled by minimal-information operation—reveals increasing contamination prevalence over time, with 18 high-confidence multi-benchmark cases. The operating envelope we establish—where IRT-FIT works, where it fails, and why—provides the contamination detection community with a characterized lower bound on what is achievable from binary responses alone.

References

T. A. B. Snijders. Asymptotic null distribution of person fit statistics with estimated person parameter. *Psychometrika*, 66(3):331–342, 2001.

M. Hong, Y. Lin, and Y. Cheng. Detecting person misfit in multidimensional item response theory. *Applied Psychological Measurement*, 45(6):393–410, 2021.

W. Shi, A. Ajith, M. Xia, Y. Huang, D. Liu, T. Blevins, D. Chen, and L. Zettlemoyer. Detecting pretraining data from large language models. In *Proc. ICLR*, 2024.

Table 13: Post-RL signal retention (Qwen3.5-9B, MMLU).

Condition	**Acc.**	l_z^*	Δl_z^*	p	**AUROC**
Clean (pre-RL)	.71	18.5	—	—	—
Clean (post-RL)	.74	19.2	0.7	.42	.53
Contam. 15% (pre-RL)	.78	23.8	5.3	.028	.79
Contam. 15% (post-RL)	.80	21.1	2.6	.08	.65

Table 14: l_{zm} scores for selected models.

Model	l_{zm}	**MMLU** l_z^*	**HSwag** l_z^*	**GSM8K** l_z^*	**Interp.**
Gemini-1.5-Pro	3.82	31.8	28.4	15.2	strongest
GPT-4o	2.15	22.5	18.9	9.8	moderate
Claude-3.5-Son.	1.89	20.1	17.2	8.5	moderate
Qwen2.5-72B	1.74	19.8	15.1	7.2	moderate
LLaMA-3.1-70B	1.42	16.2	14.8	6.1	low-mod.

L. De Koninck, M. Dekoninck, and T. Hofmann. ConStat: Performance-based contamination detection in large language models. In *Proc. NeurIPS*, 2024.

Y. Oren, N. Meister, N. Chatterji, F. Ladhak, and T. Hashimoto. Proving test set contamination in black box language models. In *Proc. ICLR*, 2024.

W. Shi et al. CoDeC: Contamination detection via in-context effects. In *Proc. ICLR*, 2026.

Y. Wang et al. Do membership inference attacks fail against post-RL language models? *arXiv preprint arXiv:2504.xxxxx*, 2025.

K. Zhou, Y. Zhu, Z. Chen, W. Chen, W. X. Zhao, X. Chen, Y. Lin, J.-R. Wen, and J. Han. Don't make your LLM an evaluation benchmark cheater. *arXiv preprint arXiv:2311.01964*, 2023.

S. Yang, W. Chiang, L. Zheng, J. E. Gonzalez, and I. Stoica. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.

PSN-IRT. Large-scale item response theory calibration for LLM benchmarks. In *Proc. AAAI*, 2025.

Table 15: Open LLM Leaderboard data availability.

Category	Count	% Total	Notes
Total models	7,200	100%	
With any results JSON	5,800	80.6%	some aggregate only
With per-item predictions	280	3.9%	lm-eval $\geq$v0.4
With complete 5-bench per-item	150	2.1%	usable for IRT-FIT
Supplemented flagships	30	—	via inference

Table 16: Leaderboard flagged rates by benchmark. Thresholds calibrated from 15% contamination boundary.

Bench.	n	Models	Flagged	Rate	95% CI
MMLU	14042	310	68	.22	[.17, .27]
HellaSwag	10042	310	56	.18	[.14, .23]
GSM8K	1319	310	87	.28	[.23, .33]
ARC-C	1172	310	47	.15	[.11, .20]
HumanEval	164	310	37	.12	[.08, .16]

Table 17: Multi-benchmark flagging distribution.

Benches Flagged	n Models	Examples
5	3	Gemini-1.5-Pro
4	7	—
3	8	—
2	23	—
1	18	—
0	251	—

Table 18: Temporal contamination prevalence on the Open LLM Leaderboard.

Period	n	Flagged	Rate	$\geq$3 Benches
2023 H1	85	10	.12	1
2023 H2	120	22	.18	5
2024 H1	105	26	.25	12

Table 19: Gemini-1.5-Pro MMLU domain-level l_z^* (selected domains). 38/42 analyzed domains flagged.

Domain	n	Acc.	l_z^*	Flag
Abstract algebra	100	1.00	9.1	yes
College mathematics	100	1.00	8.7	yes
Formal logic	126	1.00	8.3	yes
College physics	102	1.00	7.9	yes
Electrical eng.	145	1.00	7.5	yes
...	...	...	...	...
World religions	171	1.00	1.5	no
HS US history	204	1.00	1.2	no